

Assignment-3

Day-4: Component-Specific Troubleshooting Practical Identify and fix issues in faulty components (RAM, GPU, storage). Task: Create a checklist for diagnosing hardware issues

Task-1: Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

- **Shut down the PC** completely and disconnect the power cable.
- **Press the power button for a few seconds** after unplugging it to discharge remaining power.
- **Ground myself** by touching the metal chassis before handling the RAM.
- **Open the PC case** and locate the RAM slots beside the CPU socket.
- **Release the clips** at the ends of the RAM slot.
- **Remove the RAM module** by holding it by its edges. I avoided touching the gold contacts.
- **Inspect the RAM and slot** for dust, dirt, physical damage, or bent components.
- **Reinsert the RAM** by aligning the notch on the module with the key in the slot.
- **Press the module down firmly and evenly** until the retaining clips lock into place.
- **Reconnect the power cable** and start the PC.
- **Check the result:** The PC should complete POST and display normally.

Task-2: Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

- **Identify the symptom**

Check whether there is no display, screen flickering, graphical artifacts, crashes, or poor gaming performance.

- **Check the monitor and cable**

Make sure the HDMI/DisplayPort cable is connected properly.

Try another cable or monitor if available.

- **Check the GPU physical connection**

Shut down and unplug the PC.

Make sure the GPU is properly seated in the PCIe slot.

Check that the GPU's PCIe power connectors are firmly connected.

- **Check the display output**

Make sure the monitor cable is connected to the **GPU's video output**, rather than the motherboard video output when using a dedicated GPU.

- **Check the GPU drivers**

In Windows Device Manager, check out **Display adapters** for warning symbols or errors.

Reinstall/update the appropriate graphics driver if necessary.

- **Check GPU cooling**

Make sure the GPU fans can spin and that the heatsink is not heavily blocked by dust.

Check for excessive overheating during operation.

- **Test the GPU**

Run a suitable graphics test/game and observe whether the original problem occurs again.

- **Confirm the fix**

Verify that the display is stable, artifacts are gone, and the PC can run a graphics workload without crashing.

Task-3: Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Before disconnecting

1. Shut down the test PC completely.
2. Disconnect the power cable.
3. For a **SATA drive**, disconnect the SATA data cable and/or SATA power cable.
4. For an **NVMe SSD**, remove it only if the test system is designed for this and you can safely access the M.2 slot.
5. Start the PC without the drive connected.

Error/BIOS indicators

Depending on the motherboard and operating system, you may see messages such as:

- **“No bootable device”**
- **“Boot device not found”**
- **“Operating System not found”**
- The drive may be **missing from the BIOS/UEFI storage information**.

If the disconnected drive contained the operating system, the PC may fail to boot into Windows.

After reconnecting

6. Shut down the PC again.
7. Reconnect the SATA data/power cable or properly reinstall the NVMe drive.
8. Start the PC and enter BIOS/UEFI if necessary.
9. Check the **Storage/NVMe/SATA information** to confirm that the drive is detected.
10. Continue booting into the operating system.

Expected result: The drive should appear again in BIOS/UEFI and, if it contains a valid operating system, the PC should boot normally.

Task-4: Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Among the three choices of **RAM, GPU, and storage**, I would check the **RAM first**.

Reasoning

Faulty or poorly seated RAM can cause system instability, crashes, freezes, and unexpected restarts. Storage problems more commonly produce boot errors, missing files, or application/OS errors, while GPU problems often show display artifacts, driver crashes, or loss of video.

RAM Diagnostic Steps

1. **Power off the PC** and disconnect it from the power source.
2. **Reseat the RAM**
 - a. Remove the RAM modules.
 - b. Inspect the modules and slots.
 - c. Reinstall the RAM firmly.
3. **Test one RAM module at a time**
 - a. If the PC has two RAM sticks, remove one and test the PC.
 - b. Repeat with the other module.
4. **Try different RAM slots**
 - a. This helps determine whether the problem could be related to a particular motherboard slot.
5. **Check POST behavior**
 - a. Listen for beep codes or check the motherboard's diagnostic LEDs if available.
6. **Run a memory test**
 - a. Use a bootable memory testing utility such as **MemTest86** to check for memory errors.
7. **Confirm the fix**
 - a. Run the PC normally and perform a workload that previously caused the restart.
 - b. If the system remains stable and the memory test reports no errors, the RAM-related problem may have been resolved.

Important: Random restarts can also be caused by the **PSU, CPU/GPU overheating, motherboard problems, or unstable power**, so if RAM testing is normal, those components should be investigated next.

